Finding Kevin De Bruyne's Successor

Replicating Manchester City's player-similarity model with public data · Jun Jeon · September 2026

- **Question.** Among 2024/25 Big-5 midfielders and forwards, who looked most like *peak* Kevin De Bruyne (2019/20–2022/23) — and does Rayan Cherki, the player Manchester City actually signed, come out near the top?
- **Answer.** Yes. Cherki ranks **4th of 926** candidates (cosine similarity 0.954), inside a tight top group with Olise, Maddison and Isco.
- **Robustness.** Across 27 model variants Cherki ranks **3rd–8th (median 4th)** and is top 10 in all 27 — one of only four players to manage that.
- **The gap.** Cherki matches De Bruyne on attacking role and playmaking volume; the one big difference is end product — the dimension a 20-year-old is most likely to grow into.

1. Background

In the Prime Video documentary *A Beautiful Obsession* (August 2026), Manchester City's recruitment staff described how they searched for De Bruyne's successor: 43 performance metrics were condensed into 7 numbers, players were clustered, and the market was ranked by similarity to *peak* De Bruyne. Cherki, signed from Lyon in summer 2025, was the answer — followed by human checks (coach references on mentality, a video call with Pep Guardiola) before the deal. City's exact algorithm is not public. This project reconstructs the data step with public data and a standard, transparent pipeline, and tests how much the answer depends on the choices made.

2. Data and method

- **Data.** FBref player-season stats (provider: Opta), from two Kaggle snapshots taken before FBref removed its Opta advanced stats in January 2026: 2024/25 Big-5 (267 columns) and 2017/18–2023/24 Big-5 (65 columns). Both were verified to be Opta-era data; 2023/24 vs 2024/25 league medians agree within ±10% on 33 of 35 shared metrics, confirming identical definitions.
- **Features.** 28 metrics available in *both* datasets, per 90 minutes: shooting & chance creation, progression, passing, dribbling, defending, position. Expected assists (xAG) is absent from the historical data — its impact is measured in Section 5.
- **Candidate pool.** 2024/25 outfield players with a midfield or forward role and ≥ 900 minutes: **927 players** (mid-season transfers merged). De Bruyne's own 2024/25 season is in the pool as a check — it ranks 7th — but is excluded from the ranks reported below.
- **Peak De Bruyne.** 2019/20–2022/23, minutes-weighted (9,406 minutes).
- **Model.** Standardise the 28 features on the pool → PCA to 7 components (80% of the variance) → cosine similarity to peak De Bruyne in component space.

3. Result: Cherki ranks 4th

The top of the ranking is a cluster rather than a single answer: only 0.013 separates 1st (Olise) from 4th (Cherki), and Cherki is more similar than 99.6% of the pool. The rest of the top 15 — Güler, Szoboszlai, Wirtz, Simons, Ødegaard — are recognisable creative No. 8/10 profiles, which is a useful plausibility check on the model. Among players aged 22 or under, Cherki ranks **2nd**, behind Olise and ahead of Güler.

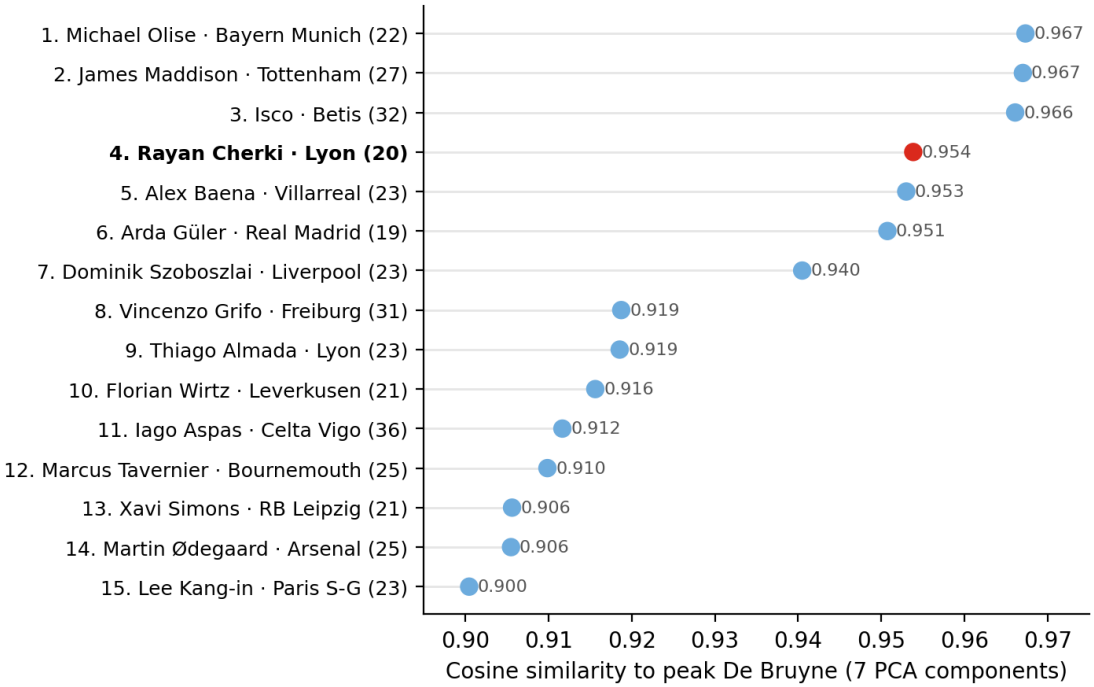


Figure 1. Top 15 of 926 candidates by similarity to peak De Bruyne (age at start of 2024/25 in brackets).

4. How robust is the result?

Three choices in the model were ours, not City's: the number of components, the minutes threshold, and the definition of *peak*. Varying all three together gives 27 variants (scaler and PCA refitted each time).

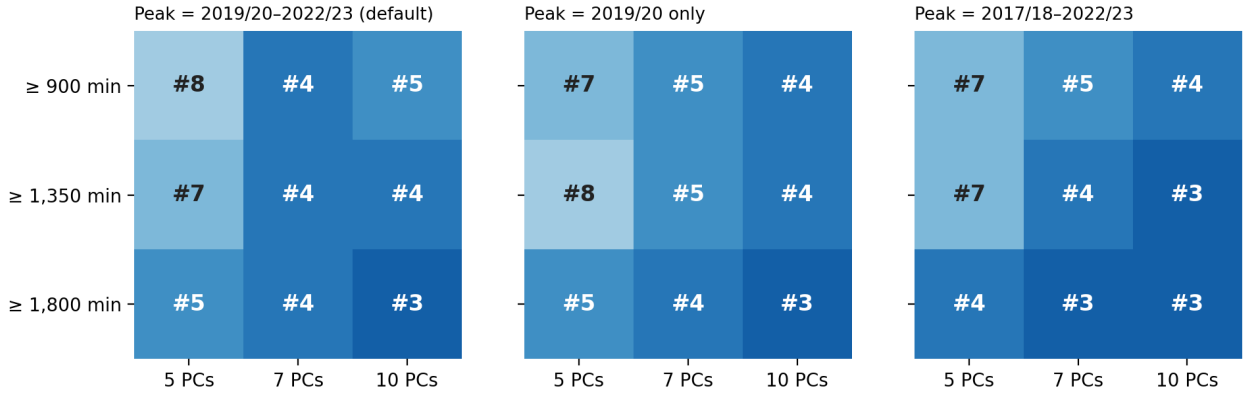
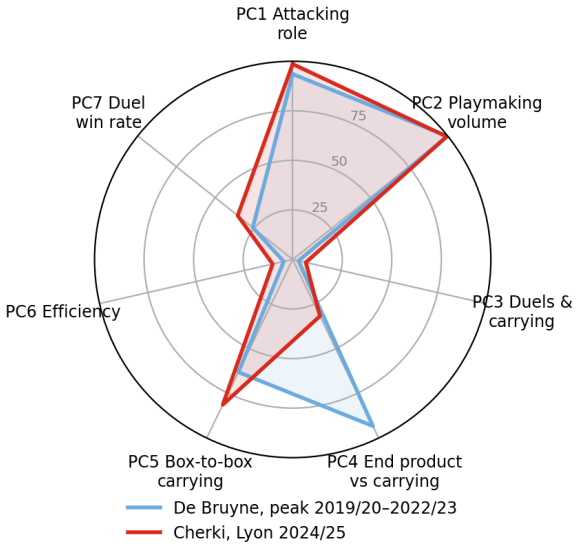


Figure 2. Cherki's rank in each of the 27 variants (darker = higher).

Cherki's exact position moves between 3rd and 8th, but he is in the top 10 in **every** variant; only Olise, Maddison and Baena match that. His lowest ranks come from the 5-component model, which discards the most detail; more components or stricter minutes filters move him up. **The robust conclusion is that Cherki belongs to a small, stable group of peak-De Bruyne profiles — not that he is precisely 4th.**

5. Where Cherki differs — and the missing xAG



Each PCA component was named from its largest loadings; the radar shows both players' percentile in the 2024/25 pool on each component.

Where they match: the two dominant components — attacking role (PC1, 99th vs 94th percentile) and playmaking volume (PC2, 99th vs 100th) — which together hold 56% of the variance, plus the minor style components.

Where they differ: PC4, *end product vs ball-carrying*. Peak De Bruyne turned his play into goals and assists at an elite rate (93rd percentile); Cherki, who carries the ball more, sits at the 32nd. That is the plausible growth area behind City's view that Cherki could become De Bruyne's profile within two years.

Figure 3. Percentile within the 2024/25 pool on each of the 7 components. Component names are working labels; PC3–PC7 each explain 3–8% of the variance.

Does the missing xAG bias the result? Peak-season xAG is unavailable, so we tested the pipeline with a reference that has it — De Bruyne's own 2024/25 season — ranking the pool with and without xAG. The rankings are almost identical (Spearman ρ = 0.996; 19 of the top 20 unchanged), and where xAG does matter it helps Cherki (4th → 2nd): his xAG per 90 (0.49) is in the pool's top 1% and above De Bruyne's 2024/25 value (0.42). Leaving xAG out therefore makes the main model, if anything, conservative about Cherki.

6. Limitations

City's model is not public	This is a reconstruction of the described process (43 → 7 → similarity), not City's algorithm. We use 28 metrics, not 43.
No expected assists	Tested above: minimal effect on the ranking, and it would favour Cherki.
No league adjustment	Per-90 output in Ligue 1 is compared directly with the Premier League; a league-strength correction is a planned extension.
Event data only	No tracking or physical data (speed, pressing intensity, off-ball runs) is freely available.
Mentality is not measurable	City's decisive checks — coach references and a conversation with the manager — are outside what data can capture.
Approximations	Rates for mid-season transfers are minutes-weighted across clubs; De Bruyne's peak mixes four seasons of different team contexts.

7. Next steps

- **Post-hoc check:** did Cherki's 2025/26 season at City move towards De Bruyne's profile, especially on PC4 (end product)?
- **Back-test:** apply the same pipeline to other replacement signings to see whether the method would have found them.
- **League adjustment:** a simple league-strength correction and its effect on the ranking.

Reproducibility. Four Google Colab notebooks (English): 01 data acquisition & checks · 02 feature selection & preprocessing · 03 similarity model · 04 robustness & interpretation. Data: Kaggle *hubertsidorowicz/football-players-stats-2024-2025* and *akshankrithick/fbref-2017-2024-for-europes-top-5-leagues* (FBref / Opta).